

Project Name	Voice Patient Diary & Clinical Integration	Maximum Marks	4 Marks
Date	15 February 2025	Team ID	6ab615fddcdae33acb38af6f

Solution Architecture

Overview: A solution architecture is a multi-tier framework that connects business requirements to a working technology implementation. It sets out scalable design patterns, security practices, and operational workflows across integrated cloud services.

Core Objectives & Goals

Technical Alignment Identify and deploy the best cloud and software solutions for the core clinical and data-ingestion problems.	Stakeholder Communication Describe the structure, behavior, and data flow of the application for developers and stakeholders.
Feature & Phase Definition Define system features, development phases, and technical requirements for systematic delivery.	Operational Specifications Provide the precise technical specifications by which the solution is defined, managed, and validated.

Solution Architecture Diagram

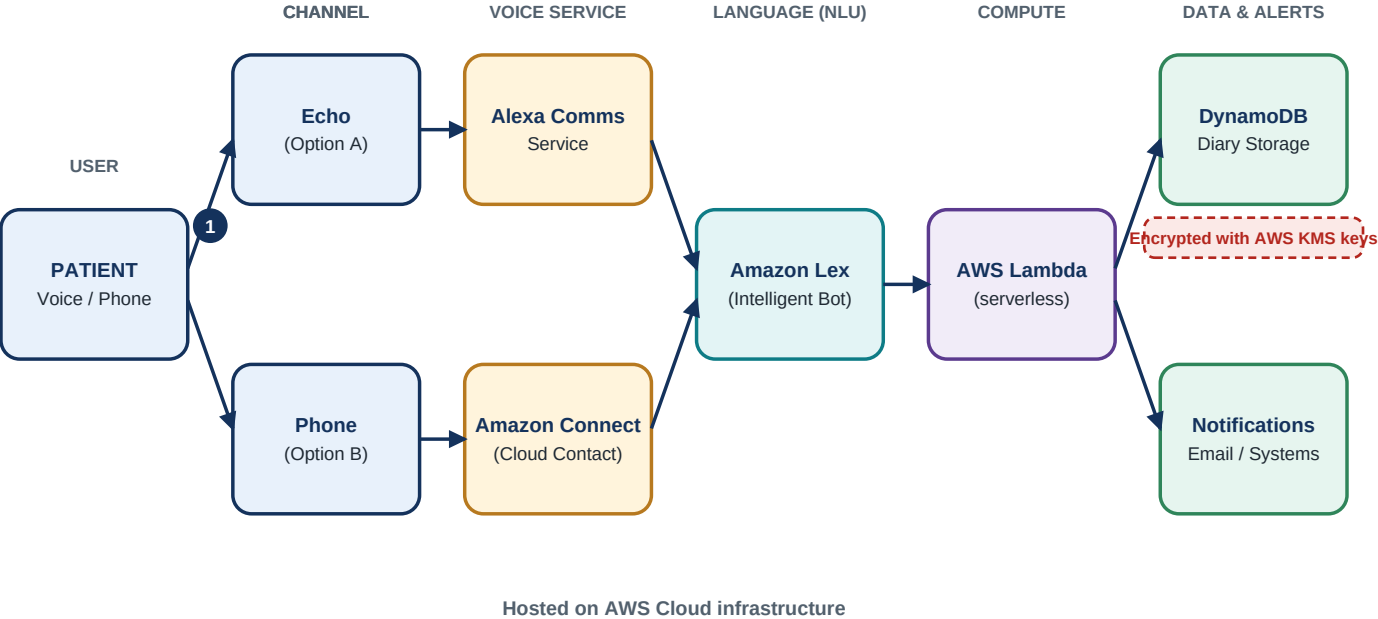


Figure 1: Architecture and data flow of the voice patient diary sample application (AWS Cloud infrastructure)

Architecture highlights: Conversational voice interfaces (Amazon Alexa and Amazon Connect) feed natural language understanding in Amazon Lex. Intent slots, such as medication logging time, are processed by serverless AWS Lambda functions and stored in Amazon DynamoDB, with encryption keys managed through AWS KMS.

Data Flow

How a patient's spoken diary entry moves through the system.

Step	Stage	What Happens
1	Patient speaks	The patient records a diary entry by voice on an Echo device (Option A) or by phone call (Option B).
2	Voice service receives	Alexa Comms Service handles the Echo path; Amazon Connect (cloud contact center) handles the phone path.
3	Intent understanding	Amazon Lex interprets the request and extracts intent slots, for example the medication logging time.
4	Serverless processing	AWS Lambda validates the slot values and applies the diary logic.
5	Secure storage	The diary entry is saved in Amazon DynamoDB, encrypted with keys managed through AWS KMS.
6	Notification	Email or other connected systems are notified so clinical teams can act on the entry.

Components and Roles

Component	Role	Service
Echo / Phone	Patient-facing voice channels (Options A and B)	Option A and Option B
Alexa Comms Service	Voice entry point for the Echo path	Alexa
Amazon Connect	Cloud contact center for the phone path	Amazon Connect
Amazon Lex	Conversational bot with natural language understanding	Amazon Lex
AWS Lambda	Serverless processing of intent slots	AWS Lambda
Amazon DynamoDB	Encrypted diary storage	DynamoDB, AWS KMS
Notifications	Email and system alerts to clinical teams	Email / connected systems

Risks and Mitigations

Risk	Mitigation
Voice recognition errors in slot values	Confirm key values back to the patient (for example the logged time) before saving.
Protected health information in transit and at rest	Keep KMS encryption on storage; use encrypted connections between services.
Compliance requirements for health data	Confirm applicable regulations and service eligibility before production use.
Failed or lost notifications	Add retry and failure alerts so clinical teams are not left uninformed.

Risks and mitigations are suggestions for the team to review and confirm.